



INSTITUTO SUPERIOR TÉCNICO

DEPARTAMENTO DE ENGENHARIA INFORMÁTICA

FORENSICS CYBER-SECURITY

MEIC, METI

Lab Assignment II

OPERATION DEEPFISH – Stage II

2026/2027

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Introduction

This second assignment will continue the investigation of the “OPERATION DEEPFISH” case. In Lab Assignment I, you were tasked with forensically analyzing the suspicious emails and the website they referenced. Your investigation revealed that the site was hosting a phishing campaign designed to capture user credentials and also uncovered a set of hidden files suggesting that a larger scheme may be unfolding. These findings provide the basis for the next phase of the investigation. Both the phishing website and the hidden files recovered during Lab Assignment I are available through the course website.

Next, you will examine new digital evidence obtained as the investigation progressed: hard disk images retrieved from the personal Windows laptop and Linux workstation of the main suspect in the case. Your objective is to determine what this evidence reveals about the phishing campaign, the previously recovered artifacts, and any other activity relevant to the investigation. You will need to develop your skills in memory and file system forensics. All required digital artifacts are available on the course website. As in the first assignment, we recommend using Kali Linux in a controlled virtual machine for your analysis.

Scenario presentation

During Lab Assignment I, your investigation of the hyperlink contained in the suspicious email led to the discovery of a cloned website designed to imitate the Fénix IST login page. Analysis of the website established that it was part of a phishing campaign intended to capture credentials submitted by its victims.

Further forensic examination of the website’s Git repository revealed eight source artifacts containing hidden files. Some of these artifacts were recovered from the repository’s commit history, while others were retrieved through Git Large File Storage (LFS). Although the precise significance of the hidden files remains unclear, their contents suggest that the phishing campaign may be connected to a broader operation.

The investigation team compiled the recovered source artifacts, their corresponding hidden files, and the associated images into a shared repository, available at: <https://drive.rnl.tecnico.ulisboa.pt/s/7F3kaKKYF22385n>

Table 1 identifies the eight original source artifacts, their SHA-256 hashes, and the hidden files associated with each artifact. Table 2 lists the recovered hidden files, their SHA-256 hashes, and a brief description of their contents. These hashes provide a reference for verifying the integrity of the material made available by the investigation team.

Table 1: Original source artifacts, their SHA-256 hashes, and the corresponding hidden files.

Source artifact	SHA-256	Hidden file
blank	23cd980099fab1c0b6993a9d27899f8679ffe42d423d8d51b29c655c0d53bbec	f1.png
polish_cow.webp	857f75db04db84340d4efa53061ca26fc7c7e6f65f80899c57f8d65b4c9d5678	f2.json
myzip.zip	9ddeef450f4307a974d07516519d076429b9bd1a32e192816ae0217cc8139de4	amelia.pdf
GrandTheftAutoVITrailer1.mp4	5d3b5f08f35c5d812e0f2c36d58c9c4f2a0b844e2e1f8bd5054a9bc54f8471ca	f4.pdf
ESP32_ADXL345_3Axis_Telemetry.xlsx	3369cd48df5bb112def25e662398674e89bdeb5df6eaa400c4ba538a9778f218	f5.py
gta_v_iceberg_unexplained.jpg	7d0f0c05c98fe80513e8e311918298f8539835d960e7270185ed25a5163d173d	f6.json
dj_cara_after_hours.wav	2b61eecffc5261dcc05d833c64454b42d33f445d3df09095988d0c2dbbc8753d	f7.diff
[Erai-raws]ShingekinoKyojin-01[720p][DualAudio].mkv	9de6f912cfaffcb8d68f4c2db39dff1e22994c92e56e8daac70e542caf67b9b7	f8.webp

Table 2: SHA-256 hashes and descriptions of the eight hidden files.

Hidden file	SHA-256	File description
f1.png	76df940819110fc996599a9df24d54c3c55eaab85055cf1306acc2c375ee396b	Reddit thread screenshot
f2.json	4cf80e2f2a1321b13e3dece45d0c949de73e5aad3ca9d1d8e4339b398101110	System tools configuration file
amelia.pdf	04be0e12533ede61c718f63b3bedf9888b2914dfe0e437d9c0f1272b24f51707	AI architecture technical document
f4.pdf	0a4ac4174b6e68c7e92912c588e67dbf351a332d201c857f592fedb0397d005f	Technical assurance audit report
f5.py	27a38aa066b3fe602c9aed0695da0d8146d28eab3e2a004cd1d0a168356db9d	Model training Python script
f6.json	9e6333a767e936973af82e68ea61023388ce6b5d8697b8652985e186a305f78f	Cloud service account credentials
f7.diff	a0fa5c55487cbf9a8e01eeb65d9bdca53770929508508093b475641c3cee1fc3	Query module code patch
f8.webp	5fb499e82ca618def24640a1e6298512646636fae243d928f10ebdbd29e50e10	Messaging chat conversation screenshot

Attribution and Seized Evidence

Since the operation of a phishing website constitutes a criminal offense, the investigation proceeded with a focus on identifying the person(s) responsible. Further analysis of the website and its hosting infrastructure revealed links to resources associated with an IST account. Investigators also identified publicly accessible GitHub and GitLab accounts using the username MuskyBoi (<https://github.com/MuskyBoi> and <https://gitlab.com/MuskyBoi>). Information associated with these resources led investigators to João Musk, an IST MSc student, as a person of interest in the investigation.

João was subsequently contacted and questioned about his possible involvement in the phishing campaign. He denied any responsibility and declined to cooperate further. The case was referred to the Polícia Judiciária and, with judicial authorization, investigators searched João's residence and seized two computers: a personal Windows laptop and a Linux workstation. Both computers were connected to the Internet through the local network at the time of the search.

The Windows laptop was powered on and running when investigators arrived. In addition to acquiring forensic images of the storage devices, investigators therefore captured the live memory of the laptop before it was seized. No corresponding memory capture is available for the Linux workstation.

During the search, investigators also recovered the handwritten note shown in Figure 1. Its relevance, if any, to the investigation has not yet been established.

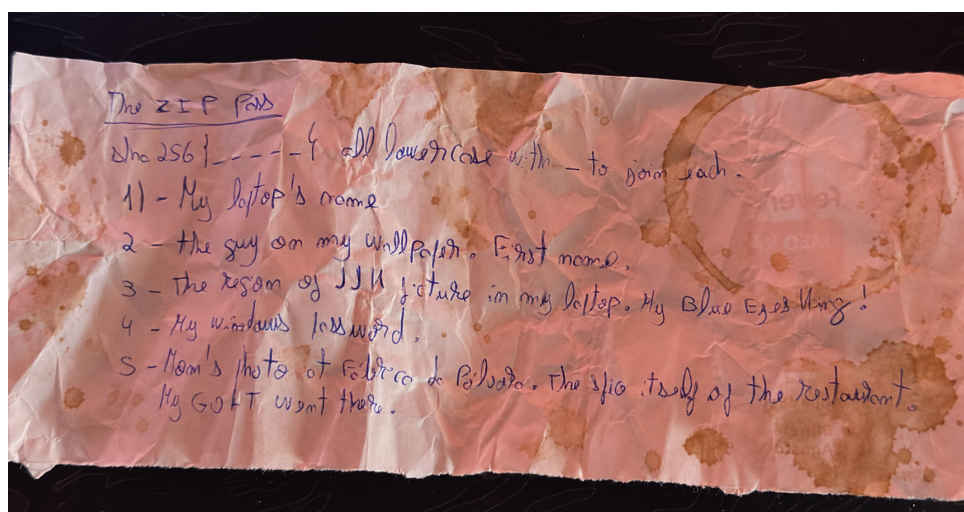


Figure 1: Handwritten note recovered during the search.

The acquired disk images and live-memory capture are available on the course website under *Course Material > Lab assignments*. The handwritten note is reproduced above and should also be treated as

part of the Stage II evidence set.

Evidence	Description
windows_laptop_disk.dd	Forensic disk image of João's Windows laptop The secondary D: data partition is BitLocker-encrypted
windows_laptop_memory.raw	Live-memory capture of the Windows laptop acquired while the system was running
linux_workstation_disk.dd	Forensic disk image of João's Linux workstation

Forensic Assignment

Following the findings of Lab Assignment I and the referral of the case to the Polícia Judiciária, your group has been invited to continue supporting the investigation by examining the newly acquired digital evidence. Your mission is to determine what the seized systems reveal about the phishing campaign, the hidden files recovered during Lab Assignment I, and any other activity relevant to the case. You should seek to establish relationships between evidence from different sources, reconstruct relevant activity over time, and identify any additional findings that may help explain the events under investigation.

Your investigation will be evaluated based on two complementary components: the forensic examination and analysis performed, and the conclusions drawn in response to the case questions.

Investigation and evidence analysis. Conduct an impartial and technically supported examination of the available evidence. Material findings must be supported by identified evidence and documented in sufficient detail for the analysis to be understood and, where applicable, reproduced. Pay particular attention to evidence provenance, integrity, and temporal context.

Questions to answer. Based on the evidence examined during the investigation, your report must explicitly answer the following questions. Clearly distinguish observations supported directly by evidence from forensic inferences and hypotheses, and identify the evidence supporting each material conclusion.

1. Do the seized systems contain traces of the phishing website and/or the hidden files recovered during Lab Assignment I? If so, identify the relevant evidence, its location, and its SHA-256 value where applicable.
2. Can you reconstruct the provenance and handling of these files? Provide a timeline of relevant events showing how the phishing website was developed and operated, and how the hidden files were handled on the seized systems.
3. Did you find evidence of anti-forensic activity? If so, describe the techniques observed and the evidence supporting your conclusions.
4. What new discoveries emerge from the Stage II evidence that were not visible during Lab Assignment I? Do these findings strengthen or weaken the attribution of the phishing attack to João Musk, or point to other relevant actors or infrastructure? Based on your analysis, what hypotheses can you formulate about the larger scheme suggested by the hidden files, and what further investigative steps would you recommend to the Polícia Judiciária?

Note: The Linux workstation forensic image provided in this exercise was created in an emulated environment. Its acquisition metadata gives 2026-09-20 as the date the disk image was created. For this exercise, however, treat the image as if it had been created on 2026-08-24. When you reconstruct timelines, subtract 27 days from every absolute timestamp you find in it. For example, a file whose recorded timestamp in the image is “2026-09-21 14:10” should appear as “2026-08-25 14:10” in your timeline. Timestamps in the Windows image need no adjustment.

Deliverables

Submit a single ZIP archive through Fenix by **7 October 2026**. The archive must contain the following four deliverables:

- **Digital forensic report.** A report documenting the investigation and evidence analysis performed in accordance with the requirements specified in the **Forensic Assignment** section. The report must explicitly answer all case questions and demonstrate compliance with the **Requirements**. Each material finding or conclusion must be supported by identified evidence and by a sufficiently documented and reproducible analysis process. Use the report template available on the course website unless instructed otherwise.
- **Evidence artifacts.** The relevant evidence artifacts recovered or preserved during the investigation. The report must identify each submitted artifact by file name and SHA-256 hash, where a stable byte-exact representation is available, and state its source or provenance.
- **Auxiliary items.** Any scripts, programs, command outputs, notes, or other supporting material required to reproduce or substantiate important findings.
- **Timeline.** A timeline of the most relevant events in the case. Each event must be supported by evidence and identify the source of its timestamp, together with any assumptions or time-zone conversions applied. Use the timeline template available on the course website unless instructed otherwise.

Use of Artificial Intelligence

The use of Artificial Intelligence (AI) tools is permitted in this assignment, provided that it complies with the applicable Instituto Superior Técnico guidelines for the responsible use of AI.

Students remain fully responsible for all submitted work, including findings, technical conclusions, scripts, commands, procedures, and any other material produced with the assistance of AI tools. AI-generated or AI-assisted content must be independently verified for correctness, validity, and relevance.

Students must understand and be able to explain, justify, and, where requested, reproduce or demonstrate the investigation they submit, including the procedures followed, the tools and techniques used, the evidence obtained, and the conclusions presented. Failure to do so may be taken into account when assessing whether the student has demonstrated the competencies evaluated by this assignment.

Any use of AI tools must be declared in accordance with the applicable Técnico guidelines, identifying the tools used and the purposes for which they were used. Students should consult the guidelines and regulations published by the Conselho Pedagógico at: <https://conselhopedagogico.tecnico.ulisboa.pt/regulamentos/faqs-regulamento-de-avaliacao-2026/>

The Fundação para a Ciência e a Tecnologia (FCT), through FCCN, also provides the **IAedu** platform to the Portuguese higher-education and research community. The service is available to Técnico students using their institutional credentials and provides access to several AI models without requiring an individual paid subscription. Students wishing to use AI tools in this assignment are encouraged to use this institutional service: <https://si.tecnico.ulisboa.pt/servicos/servicos-externos/iaedu/>

Good luck!